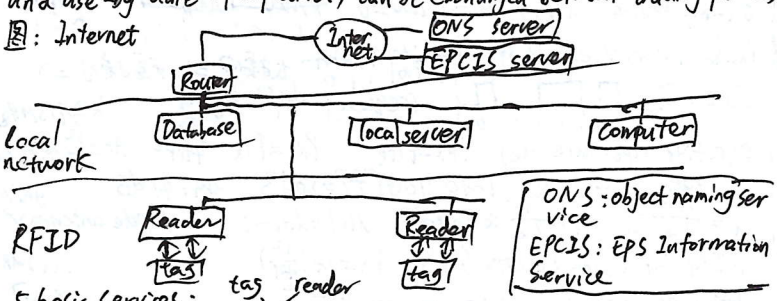
RFID: is a technology that uses radio waves to transfer energy and data between a reader and a movable item to identify, track, RFID is fast, reliable, doesn't require physical sight or contact
组成: tags (transponder): store data 分为 passive & active tags
reader: transmit and receive signals between reader and tag 分 Fixed/hand held
antenna: allow data exchange between tag and reader
How rfid work? reader send signal to tag, which powers it cause tags to respond with its stored data. Tag emits unique ID or data. data sent back to reader, transmits for processing.
unlike barcode, rfid 不用 direct line of sight between reader and tag.
Automatic ID system: Barcode、OCR (optimal character recognition)、RFID、smart cards、Biometric MM (包含 fingerprint、voice identification)
① Biometric procedures: science of counting and body measurement
· Fingerprint: uniqueness and stability to verify (high speed, security, low price)
· Speech recog: Feature extraction → pronunciation dictionary → Statistical acoustics model → language model → decoder
technical feature: easy to use, especially accepted by user; cheap sound equipment, not involve privacy; Due to lack of standards, some difficulties in promoting → based on human face features. analyze face images.
· Face recog: Face acquisition → face positioning → face recognition preprocessing → look up the identity → identity confirmation
Technical features: surroundings, hair ornament, age can affect the rate, fast speed of recog, not easy to be aware of it.
· Barcode: binary code compromising of field of bars and gaps arranged in parallel widely used until UPC was introduced in 1973
most common barcode by some margin is EAN code, designed in 1976.
1D barcode: combination of lines and blanks in accordance with certain coding rules, only express infor in the horizontal direction, vertical no infor.
特点: limited storage capacity, need to combine with database. The size is relatively large, space utilization is low. Fault tolerance is poor.
2D barcode: use black and white graphics on the 2-D plane to record data, use 3 positioning points for reader to identify. fault tolerance better
特点: large capacity, high infor density; support for encryption (multiple features)
Card Tech: magnetic/Smart/Optical cards RFID 识别范围: 0-5m
RFID vs barcode: more efficiency/durability (condition)/security (encrypt)
In 2003, EPCGlobal started to be standarlize RFID; RFID patented from 1973
1990s: EPC introduce; dates back: started as early as World War II, airplanes ID
main features of RFID: ① Non-contact automatic and rapid identification ② Permanently store a certain amount of data ③ Simple logic processing ④ Reflection signal strength is affected by distance ⑤ low cost, large scale
RFID challenges: limited tag resources; Performance in a real environment
energy acquisition mode: traditional wired/Battery/Wireless/Backscatter
RFID Makeup? Reader + Tags + Antenna + Middleware + RFID software + computer + DB
Reader function: connect application layer and tags, 作用如下:
· Energy supply to tags · communication · Security Assurance
分类: LF & HF: <1m, 125kHz, 135kHz, 6.78MHz & 13.56MHz (time standard, Amateur) 2.4/5.8GHz
UHF: 300M ~ 3GHz (cellphones, WLAN) SHF: 3G-30GHz (Satellite、5G mobile)
Fixed (wired, fast set up)、Portable (small, easy to move)、Industrial (other sensor)
RFID reader block diagram
upper-computer ⇄ Data ⇄ Signal Processing & Control (baseband control) ⇄ send signal / receive ⇄ Radio Frequency Module (high frequency interface module) ⇄ Antenna
RF Module 2 types: Inductively coupled RF Modules / Eletriomagnetic Backscatter Coupled RF Module 前一个适用于 short range (magnetic field) 后面 long range
Reader antenna: Omnidirectional ante. (lower gain) ② Parabolic antenna ③ half-wave dipole ante./dipole/folded dipole/printed dipole
Tag function: data storage/Contactless R&W/Energy harvesting/security encryption collison/
分类: card-like/label-like/Implantable/Accessories-like or low/High/UHF tags
low (30k-300k) H (300k-30M) UHF (300M-3GHz) or Read-only/R&W or active tags & passive tag /semi-passive tags; passive: no battery; active tag energy
tool transponder (有 power)、movement、expensive
Tag component: antenna + chip
antenna: a conductive structure designed to couple or radiate electromagnetic
分类: coil/Microstrip patch/Dipole antenna (used for UHF tag)
tag chip: is an integrated circuit in silicon consist of RF front end, a control unit
$f=\frac{1}{2\pi\sqrt{LC}}$ $C=\frac{1}{(2\pi f)^2 L}$ (C) $1pF=10^{-12}F$ $1mH=10^{-3}H$ (L) $\lambda=\frac{c}{f}$ $c=3\times10^8 m/s$
Middleware: one of the primary benefits of using RFID mid-- is that it standardizes ways of dealing with the flood of information raw
3 motivations: providing connectivity with readers; processing ~~data~~ RFID observations; providing an application-level interface to manage readers and capture ~~data~~ filtered RFID events
$T_N=(T_{ct}+T_D)N_{tag}$ LF: 30k~300k HF: 3M~30M UHF: 300M~3G

$P_{av}=\frac{V_0^2}{2R}$ $G_{dB}=10\lg\frac{P_2}{P_1}$ $dBm=10\lg\left(\frac{P}{1\times10^{-3}}\right)$ $1mW=0dBm$
$0dBm+30dB=30dBm=1W$
On-off keying: high power=1 low=0
Pulse interval encoding: 长间隔=1 短间隔=0
Fsk: frequency shift keying
effective aperture (Ae) $P_{RX}=P_{Ae}$ $P_{RX}=P_{TX}\frac{Ae}{4\pi r^2}$ $Ae=\frac{\lambda^2}{4\pi}\approx86cm^2$
1m path loss = 32dB $1\sim10:10\lg\left(\frac{10}{1}\right)^2=20dB$ 3m: 41 dB
$G=\frac{4\pi}{\Omega_{beam}}$ $G\times\frac{4\pi}{1.6}\approx8\sim9dB$ $dBd=dBi-2.2$ dipde antenna
$EIRP=P_{Tx}(dBm)+G_{Tx}(dBi)$ $Ae=G_t\left(\frac{\lambda^2}{4\pi}\right)$ (4.2dB)
Friis equation: $P_{Rx}=P_{Tx}G_{Tx}G_{Rx}\left(\frac{\lambda}{4\pi r}\right)^2$
Signal transform: fase=0° $V_t=V_{dir}+2\times\frac{V_{dir}}{3.2}\approx1.63V_{dir}$ $\frac{P_{to}}{P_{dir}}=1.63^2\approx2.7$
vertical Polarization 更平滑
计算题: $\lambda=0.33m$ ($f=915MHz$), $P_{Tx,reader}=29dBm$, $P_{active}=-14dBm$, Reflection: $\frac{1}{2}$ of received (-3dB) Reader sensitivity = -70dBm
① Max Path loss $PL=P_{Tx,r}-P_{active,tag}=29+14=43dB$
② Max distance tag & r $r=\sqrt{\frac{P_{Tx,r}}{P_{Rx,r}}\times\frac{\lambda^2}{(4\pi)^2}}=\sqrt{\frac{86\times10^{-9}}{4\pi\times10^{-4.3}}}=3.7m$
③ actual reflection attenuation: $P_{Tx,tag}=P_{Rx,tag}-3=-14-3=-17dBm$
④ reader receive: $P_{Rx,reader}=P_{Tx,tag}-PL=-17-43=-60dBm$
⑤ using isotropic antenna at reader $G_{reader}=4$ (6dBi)
find new distance: $r'=r\sqrt{G}=7.4m$
⑥ what's new reader receive power $P_{Rx,rea}=P_{Tx,tag}\cdot\frac{G_r\cdot\lambda^2}{(4\pi r)^2}=-60dBm$
$P_{REC}=P_{Tag}+G_R+G_T-\sum losses-FSL-E_T$, $r=10^{\frac{FSL-31.75}{20}}$
$E_{dB\mu V/m}=20\lg(E_{V/m}\times10^6)$
EIRP (dBm) = $10\lg\frac{(EIRP_{(W)}\times1000)}{}$
Transmitter 33dBm → Cables and Connector Losses -3dB → Reader Gain +6dB -3dB → Free Space Loss -47dB → Tag antena gain +1dB -1dB
→ Tag Sensitivity Threshold -14dB
$P_{Tag}=P_R+G_R+G_T-\sum losses-FSL$ $\lambda=\frac{3\times10^8}{f}$
915MHz: $r=10^{\frac{FSL-31.75}{20}}$
$P_{TX}=29dBm\Rightarrow29=10\lg\left(\frac{P_1}{10^{-3}}\right)$ $P_1=10^{5.9}$, 逆向 $P_2=10^{1.6}$ $r=\frac{1}{4\pi}\sqrt{\frac{10^{5.9}}{10^{1.6}}}=3.7m$
~~$P_{REC}=\frac{P_T G_T G_R^2 \lambda^4 E_T}{}$~~
LRC: 所有字节一起异或, 结果为0!
CRC: high reliable/can't correct bit
生成: $X^4+X+1=10011$, 最高n次方, 先在被除数后补n个0, 这里4个
NRZ coding: 1 high 0 low
Manchester:
Unipolar RZ: 1:1 0:-
DBP (differential): 1: 变 0: 变回原来状态
Miller: 0: 持续不变 1: 持续改变
Modified Miller
Differential coding 1: change 0: unchange
PPC i: pause t 0: pause 2t
逆变0要反转
$f_{CR}=\frac{|f_1+f_2|}{2}$ pure
modulation: ASK、FSK、PSK
$m=\frac{a_{dm}}{a_m}=\frac{u_2-u_1}{u_2+u_1}$ ie. 4% 36.8%
conflicts among reader: TDMA、FDMA、CSMA | tag-driven: ALoHA、BT
Anti-collision: SDMA、TDMA、FDMA、CDMA | reader-driven: polling、QT、Binary Search
ALOHA: $S=Ge^{-2G}$ (pure) $S=Ge^{-G}$ (slotted) 可导致 starvation
Dynamic frame slotted ALoHA: 解决 star-- by adjust frame length
Binary Search Tree: $L(N)=\log_2N+1$
Binary Query Tree: 从0开始递增枚举, 1、00、01、000 直到有 Silent (001 silent 下一个为01)
ISO/IEC 11784、11785、14223 EPC started at 1999
In 1999, EPC starts Auto-ID-Center develop necessary tech and network
Goal: create a cost-efficient, global and open-source system / Develop less complex original standards for communication / can be implemented more cost-efficiently than existing RFID standards.
partnership 大学: Fudan university in China, KAIST in korea, The University of Cambridge in UK
In 2003, European Article Number (EAN) and Uniform Code Council (UCC) founded EPCGlobal, a worldwide, open nonprofit organization
mission: develop and oversee the standard of EPC by RFID tech. / Provide registry service of code / Coordinate with ISO for their review
EPC (electronic Product code) = domain manager num + object class num + serial num. Version num

	Version num	Domain	Object	Serial	EPC 96 Type I:
EPC 64	2	21	17	24	0l_0000 489_0006F_
	3	15	13	34	0000169DC0
EPC 96 I	2	26	13	23	
	3	28	24	36	
EPC 256	1	32	56	60	
	2	64	56	128	
	3	128	56	64	→ bits (16进制)

EPC Global network is a tech that allows trading partners to document and determine the location of individual goals. additional: production and use-by date of a product / can be exchanged between trading partners

图: Internet



5 basic services:
① EPC ② Identification system ③ EPC global middleware ④ Discovery service (DS) ⑤ EPCIS

EPC Global classes:
class 0: passive, read only transponder with 64/96 bit EPC — use binary tree based variant algorithm
class 1: passive transponders, write once read many with 64/96 支持 kill 命令, 可选保密位保护 user memory) ~~lock & kill 命令~~ / circuit
class 2: passive tag above: extended tag ID, user memory
class 3: semi-passive tag: integral power source, integrated sensing
class 4: active tag, tag-to-tag 通信, EPC Global UHF
EPC C1G1: passive tag, support kill & lock commands
ping ID: a mask filter is provided that specifies the starting position
EPC C1G2: 860~960MHz half-duplex 2个link: tag-reader 反向来

Reader ⇄ Tag (ASK / ASK/PSK) allow one reader or one tag to send signal in a single transmission

Tag FSM (7 states)
ready → Arbitrate → reply → acknowledged → open → secured → killed
power off; inquiries 得ack; optional password; ≤4 readers; no longer respond to any command

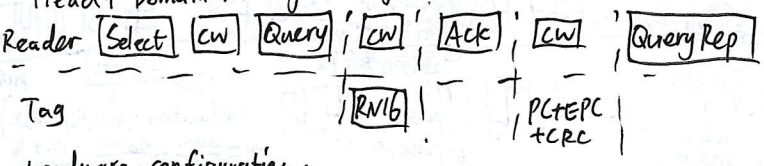
C1G2 use dynamic frame slotted ALOHA

read tag:

Session		电源开	电源关
S0 ~ S3	S0	infinite	none
	S1	2nf	500ms<t<5S
	S2	2nf	t>2S
	S3	2nf	t>2S

① select: define num of tags in next query process (22 bits)
② Query: initiate a new identify process; specify SL flag and value
③ QueryRep: indicate tag has entered next time slot. 减own slot cnt
④ QueryAdjust: used to adjust num of time slots −1, when =0 Send RN16
Q=x, 有 2^x slot 分配; SeL=10: 仅SL=0 tag 答; SeL=11 仅SL=1答

GID-96 contains 4 field
Head + Domain manager + object class + Serial number



hardware configuration:
sol1: 在middle grid 优: strong robustness; 缺: high cost, large deployment cost
sol2: each side 优: reduce hardware, reduce potential interface; 缺: lack of robustness

Software configuration: reader power / initial length of frame / coding and modulation of tags / user information of tags
external envir: energy absorption / multipath effect / polarization losses / impedance mismatch

application patterns of RFID:
① tag identification
② Information retrieval and integration
③ Object detection and tracking

① : Best setup
Tag: passive, Read only or R&W, HF or UHF — library books / ID cards — inventory
reader: fixed or Portable, HF or UHF

② Tag: passive/active(real-time), R&W, HF or UHF
reader: fixed or Portable, HF or UHF
HF used in water/metal UHF: logistics, supply chain

③
tag: passive/active, R&W, UHF ↗ active real-time location (RTLS)
reader: Fixed, UHF passive: zone-based tracking

Applications:
① Logistics — warehousing / Intelligent Transportation system / product monitoring
② smart poster / electronic business card /
③ RFID personnel real-time positioning / Medical apps to control activity range /

NFC: short-range wireless com. <20cm
use: check-in / mobile payment / fast match

security attack on transponders:
① permanent destruction of transponder
② tran — shielding / tuning
③ spoofing and cloning of transponders.

Ghost read 的 solution: Gen 2 provides 5 sequential tests that serve as checks for a tag's validity
1. tag response time 2. Preamble 3. EPC format check 4. Bit match 5. CRC

Advantages of RFID: reusable; can be read simultaneously; low cost; can be deployed in large-scale; strong penetration; physical characteristics of traditional Wireless Tech; high confidentiality

EPC C1G1 & EPC Global class 0 不 compatible (not incompatible)